ZHE FU

Personal Website: fu-zhe.com ◇ Google Scholar ◇ LinkedIn ◇ zhefu@stanford.edu ◇ Stanford, CA

RESEARCH INTERESTS

Methodology: Physics-Informed Learning, Control, Optimization, Agentic AI, LLM Interpretability

Applications: Mixed-Autonomy Systems, Autonomous Scientific Discovery, Robotic and Multi-Agent Systems

PROFESSIONAL EXPERIENCE

Academic Experience

Stanford University

Stanford Energy Fellow | Advisors: Eric Darve & Marco Pavone — Sep. 2026 -
Mechanical Engineering (ME) | Aeronautics and Astronautics (AA)
Institute for Computational & Mathematical Engineering (ICME)
Granted three years of independent research funding (over $300k), with flexible duration

Lawrence Berkeley National Laboratory (LBNL)

Berkeley Lab Affiliate | Energy Analysis Division — Nov. 2026 - Present

Industry Experience

Research Intern, Honda Research Institute, Ann Arbor, MI, USA — May 2023 - Aug. 2023
Worked on multi-agent robots collision avoidance in social navigation

Research Intern, Tsinghua Berkeley Shenzhen Institute, Shenzhen, China — July 2018 - Aug. 2018
Worked with Prof. Yi Zhang on Global Competence and Leadership Survey

EDUCATION

University of California, Berkeley

Ph.D., Transportation Engineering (CEE) | Advisor: Alexandre Bayen — Aug. 2020 - Aug. 2026
Thesis: *Physics-Informed Learning and Control for Mixed-Autonomy Systems: Enabling Traffic Smoothing with a Few Automated Vehicles*
Minors: Optimization and Networks, Data Science

M.S., Electrical Engineering and Computer Science | Advisors: Alexandre Bayen & Claire Tomlin — Aug. 2022 - Aug. 2026
Thesis: *From Density to Speed: Velocity-Domain Traffic Modeling with Neural Finite-Volume Methods*

M.S., Transportation Engineering (CEE) | Advisors: Alexandre Bayen & Xiao-Yun Lu — Aug. 2018 - May 2020
Affiliations: Berkeley's Artificial Intelligence Research (BAIR) | Institute of Transportation Studies (ITS)

Tongji University

B.S., Transportation Engineering | Advisor: Xiaohong Chen — Sep. 2014 - July 2018
Awarded National Scholarship (Top 1%)

Fudan University

Minor in Bachelor of Law (*10 courses, 30 units in total*) — Sep. 2015 - July 2017

SELECTED AWARDS & HONORS & GRANTS

Awards:

Research

- Rising Star in Data Science, University of Chicago — Nov. 2026
- Finalist, IEEE ITSC YP & WIE 3-Minute Thesis Competition — Sept. 2026
- **1st Place Winner**, H2H8 YouTube Research Presentation Competition - ***sole*** *winner* — May 2026
- **Outstanding PhD Student of the Year**, ITS Berkeley — May 2026
- Rising Star in Electrical Engineering & Computer Sciences, MIT — Oct. 2025
- **2nd Place Winner**, Berkeley Grad Slam - *Campus-wide* ***sole*** *winner* — April 2025
- Rising Star in Cyber-Physical Systems, NSF — Mar. 2025
- Rising Star in Mechanical Engineering, CMU — Oct. 2024
- **1st Place Winner**, INFORMS Research Poster Competition - *Awarded 4 out of 200, Top 2%* — Oct. 2023

Leadership

- **Eno Fellow**, Eno Center for Transportation - *Awarded to 17 graduate students nationwide* — June 2025
- Athena Award for Graduate Student Leadership, EDGE in Tech - ***sole*** *winner worldwide* — May 2025
- Dean's Award for Inclusive Excellence, Oski Student Leadership Award - ***sole*** *awardee campus-wide* — April 2024

Teaching & Mentoring

- Outstanding Graduate Student Peer Mentor Award - *Awarded to 4 students per year at Berkeley* March 2026
- Evergreen Award for Undergraduate Researcher Mentoring, EECS department at UC Berkeley May 2024
- Outstanding Graduate Student Instructor, UC Berkeley - *Top 10% among all GSIs campus-wide* March 2022

Scholarships:

- CTF Lipman Family Sustainable Transportation Graduate Fellowship - *Awarded to 4 students nationwide* Feb. 2026
- Bridgette Beato Leadership Legacy Graduate Scholarship - ***sole*** *awardee at SF Chapter* Feb. 2026
- Rodney E. Slater Scholarship - ***sole*** *awardee among Eno Fellows* June 2025
- Asian American Architects and Engineers Foundation (AAa/e) Graduate Scholarship (1st Place) Oct. 2024
- ITS California and California Transportation Foundation Scholarship Aug. 2023
- Chinese National Scholarship - *Top 1% among all undergrads* Sept. 2017

Grants:

- Stanford Energy Postdoctoral Fellowship (3 years independent funding, over $300k), *Stanford* Jan 2026
- Berkeley's Artificial Intelligence Research Lab (BAIR) Graduate Grant ($12k), *BAIR* May 2024
- Hearts to Humanity Eternal (H2H8) Graduate Research Grant ($10k), *H2H8* July 2023
- NSF Student Travel Grant, *CPS-IoT Week U.S. NSF* March 2023
- Student Research Grant ($15k), *CEE Department UC Berkeley* June 2022

TEACHING & MENTORING EXPERIENCE

Graduate Student Instructor, UC Berkeley
CE262 Analysis of Transportation Data (Graduate Level) Aug. 2021 - Dec. 2021

- Enrollment: 57
  - PhD/Master **core** course for SYS and TRANS students in CEE department
  - Students' degree programs include Undergraduate, MEng, MS, MBA and PhD
  - Students' departments include CEE, CED and Haas School of Business.
- Ratings: 4.67/5.00 (department average: 4.41, response ratio: 77.19% | 44/57)
  - Helpfulness 4.84/5.00 (avg: 4.57); Communication 4.77/5.00 (avg: 4.48)
  - Inclusive environment 4.79/5.00 (avg: 4.49); Accessibility 4.86/5.00 (avg: 4.67)
- Awarded *Outstanding Graduate Student Instructor* (Top 10%)
- Invited 3 times (2023/2024/2025) to serve as one of three panelists at International GSI conference held for around 800 first-time GSIs at UC Berkeley.
- Invited to serve as the panelist for the pedagogy class for new GSIs (CE375)

Lecturer, UC Berkeley
BAIR-NCKU Summer AI Workshop (Graduate Level) July 2025 & Aug. 2026

- Enrollment: 6 in 2025 and 7 in 2026 (Masters Students from NCKU)
- Served as the main student coordinator for the week-long module on "AI & Transportation".
- Instructed the cohort on "Learning and controls in Mixed Autonomy Traffic Systems" and "Neural Finite Volume Methods".

Lecturer, UC Berkeley
Girl Scout Engineering Day May 2023

- Enrollment: 60 (Elementary/Middle School Students from Girl Scout)
- Led a hands-on programming workshop for over 60 elementary and middle school girls, teaching basic coding principles.
- Mentored and inspired a diverse group of young girls to build confidence and pursue future careers in STEM through an engaging introduction to university-level engineering.

Volunteer Teacher, Lianhua Village School
Basic Health and Hygiene Jan. 2016 - Feb. 2016

- Enrollment: 15 (Elementary School Students in "Hui" minority)
- Volunteered to teach elementary school students at a religious school located in Lianhua village, a less-developed corner of China's mountainous west.
- Shocked by the poor health resources the left-behind children have, I designed the "Basic Health and Hygiene" course on my own, including creating the syllabus, preparing the materials and delivering 10 one-hour lectures.

Guest Lecturer, CMU & UCLA
CMU 12787 Special Topics: Physics-Informed Machine Learning April 2026

- Enrollment: 20 (Graduates: masters and Ph.D. students)

- Instructed on physics-informed learning and control for mixed traffic systems.

UCLA CEE C181/C281 Traffic Engineering Systems: Operations and Control (Undergrad & Graduate Level) Nov. 2025

- Enrollment: 30 (Undergrads & Graduates)
- Instructed on the theory and application of modeling and control in mixed-autonomy traffic systems.

Mentor, UC Berkeley
Awarded *Outstanding Graduate Student Peer Mentor Award* (Among 4 awardees campus-wide)
and the Evergreen Mentorship Award for Undergraduate Research.

Undergraduate Students (14 total; → indicates first placement after mentorship)

CEE Scholars research mentorship program (for undergrads from PREP and T-PREP programs) Spring 2024

Sami Seyedjafar Kashi, CEE, Spring 2024

BAIR undergraduate mentoring program Oct. 2023 - Present

Atharva Gupta, EECS, 2023 - 2024
Nicole Han, EECS & DS, 2023 - present → Software Engineer, X

Mobile Sensing Lab undergraduate mentoring Oct. 2021 - Present

William Tao, EECS, 2026 - present
Bhavya Mamnani, Mechanical Engineering, 2026 - present
Jordan Kam, Aerospace Engineering, 2025 - present → Ph.D. Student, Caltech Aerospace
Qing Lyu, exchange student from Tongji U, 2025 - present → Ph.D. Student, UC Berkeley CEE
Zakaria Baba, research intern from École Polytechnique, 2025 → Ph.D. Student, Caltech CMS
Martin Drieux, research intern from École Polytechnique, 2025 → M.S. Student, Carnegie Mellon University MSML
Alexi Canesse, research intern from École Normale Supérieure de Lyon, 2025 → Ph.D. Student, École Polytechnique
Xiaoyang Cao, exchange student from Tsinghua U, 2024 - present → Master's Student, MIT TPP
Jiaying Yang, exchange student from Tongji U, 2023 - 2024 → M.S. Student, UC Berkeley CEE
Ashwin Dara, EECS, 2023 - 2024 → Software Engineer, Doordash
Eric Cheng, EECS, 2021 - 2022 → Software Engineer, Unity

Graduate Students (9 total)

EECS MEng capstone project mentor Sep. 2023 - May 2024

Alvin Bao, EECS, 2023 - 2024 → Software Engineer, Netflix
Tony Luo, EECS, 2023 - 2024 → Software Engineer, Doordash
Abirami Sabbani, EECS, 2023 - 2024 → Software Engineer, Walmart Global Tech
Cathy Wang, EECS, 2023 - 2024 → Software Engineer, AWS

CEE M.S. student mentor Oct. 2020 - Present

Yichi Zhang, CEE, 2026 - present
Xuanmian He, CEE, 2025 - present
Zihe Liu, CEE, 2024 - present → Ph.D. Student, UC Berkeley
Juanwu Lu, CEE, 2021 - 2022 → Ph.D. Student, Purdue University
Jianxuan Yin, CEE, 2020 - 2021 → Construction Technology Engineer, XL Construction

LEADERSHIP & SERVICE

Leadership & Outreach

- Junior Researcher Engagement Officer, REproducible Research In Transportation Engineering (RERITE) Working Group July 2025 - Present
- Student Lead Coordinator, BAIR-NCKU Summer AI Workshop July 2025
- Initiator, IEEE Technical Committee: Automated Mobility in Mixed Traffic Mar. 2025 - Present
- Organizer, ITSC Workshop "Automated Mobility in Emerging Mixed Traffic" Series, IEEE ITSS 2024 & 2025 & 2026
- Sole Graduate Representative in CEE DEIB Committee, UC Berkeley Aug. 2023 - July 2024
- **Founder** of Representation of Asian and Pacific Islander in the Department of Civil and Environmental Engineering (RAPID-CEE), UC Berkeley
- President of RAPID-CEE, UC Berkeley May 2023 - Aug. 2024
- Graduate Assembly Delegate, UC Berkeley Aug. 2022 - Aug. 2024
- Funding Committee Member of Graduate Assembly, UC Berkeley Aug. 2022 - Aug. 2024
- Executive Board Member of Women in Computer Science Engineering (WiCSE), UC Berkeley Aug. 2022 - Aug. 2023
- Vice president of the Student Union, Transportation Engineering School, Tongji University July 2016 - July 2017

- Organizer, the **first** Asian and Pacific Islander (API) Awareness event in CEE department. Feb. 2023
- Organizer, the **first** API Social event in CEE department. May 2023
- Organizer, Berkeley-Stanford Meetup for Women in EECS. April 2023
- Volunteer, Girl Scouts Engineering Day. May 2023

Professional & Campuswide Service, UC Berkeley CEE & EECS

- Panelist, Path to the Professoriate (P2P) Program, UC Berkeley Nov. 2025
- PhD Coordinator, ITS Berkeley Seminar Series Sep. 2024 - May 2025
- Panelist, International Graduate Student Instructor (GSI) conference, UC Berkeley Aug. 2023 & 2024 & 2025
- Core member, CEE Faculty Search Student Committee 2023
- Member, EE Faculty Search Student Committee 2023
- Member, CS Faculty Search Student Committee 2023
- Volunteer, admitted students visit days in EECS and CEE. 2020 - Present
- Volunteer, TU Delft Department of Transport and Planning Visit. April 2023
- Panelist, pedagogy class for new GSIs in CEE (CE375) April 2022

PUBLICATIONS

Journals (Published & Under review) * Equal first author, † Corresponding author

J1. "Supervised and Unsupervised Neural Network Solver for First Order Hyperbolic Nonlinear PDEs." [link]
Submitted to *SIAM Journal on Scientific Computing.*
Zhe Fu*†, Zakaria Baba*, Alexandre M. Bayen, Alexi Canesse*, Maria Laura Delle Monache, Martin Drieux*, Nathan Lichtlè*, Zihe Liu, Hossein Nick Zinat Matin*, Benedetto Piccoli.

J2. "Energy-Saving Control of Freeway Traffic: Field Experiments and Results."
Submitted to *Nature.*
Sean McQuade, ..., **Zhe Fu**, ..., Jonathan Sprinkle. [All researchers in MegaVanderTest]

J3. "Safety-Critical Scenario Generation via Adversarial Imitation in Mixed-Autonomy Traffic."
Submitted to *IEEE Transactions on Intelligent Transportation Systems.*
Qing Lyu, **Zhe Fu†**, Alexandre M. Bayen.

J4. "Validation and Calibration of Energy Models with Real Vehicle Data from Chassis Dynamometer Experiments." [link]
Submitted to *Vehicle System Dynamics.*
Joy Carpio, Sulaiman Almatrudi, Nour Khoudari, **Zhe Fu**, Kenneth Butts, Jonathan Lee, Benjamin Seibold, Alexandre M. Bayen

J5. "Hybrid Multi Class Traffic Flow Solver."
Submitted to *Transportation Research Part C.*
Diego Guaman, Juan C. Herrera, Hans Lobel, **Zhe Fu**, Alexandre M. Bayen

J6. "Kernel-based Planning and Imitation Learning Control for Flow Smoothing in Mixed Autonomy Traffic." [link]
Accepted at *Transportation Research Part C*, 2024.
Zhe Fu†, Arwa Alanqary, Abdul Rahman Kreidieh, Alexandre M. Bayen.
- **First-place Winner**, INFORMS Annual Meeting Poster Competition (4 out of 200), 2023.
- Honorable Mention. OR/MS Tomorrow Mini-poster Competition (PhD Category), 2023.
- Accepted by 25th International Symposium on Transportation and Traffic Theory (ISTTT25), 2024.

J7. "Hierarchical Speed Planner for Automated Vehicles: A Framework for Lagrangian Variable Speed Limit in Mixed Autonomy Traffic." [link]
Accepted at *IEEE Control Systems Magazine*, 2024.
Han Wang, **Zhe Fu**, Jonathan Lee, Hossein Nick Zinat Matin, Arwa Alanqary, Daniel Urieli, Sharon Hornstein, Abdul Rahman Kreidieh, Raphael Chekroun, William Barbour, William A Richardson, Dan Work, Benedetto Piccoli, Benjamin Seibold, Jonathan Sprinkle, Alexandre M. Bayen, Maria Laura Delle Monache.

J8. "Traffic Smoothing via Connected & Automated Vehicles: A Modular, Hierarchical Control Design Deployed in a 100-cav Flow Smoothing Experiment." [link]
Accepted at *IEEE Control Systems Magazine*, 2024.
Jonathan Lee, ..., **Zhe Fu**, ..., Alexandre M. Bayen. [All researchers in CIRCLES consortium]

J9. "Evaluating Smart Charging Strategies using Real-world Data from Optimized Plugin Electric Vehicles." [link]
Accepted at *Transportation Research Part D*, 2021.
Sierra I. Spencer, **Zhe Fu**, Elpiniki Apostolaki-losifidou, Timothy E. Lipman.

J10. "Economic and Environmental Benefits for Electricity Grids from Spatiotemporal Optimization of EV Charging." [link]
Accepted at *Energies*, 2021.
Soomin Woo, **Zhe Fu**, Elpiniki Apostolaki-losifidou, Timothy E. Lipman.

Journals (In Preparation)

J11. From Density PDEs to Velocity Dynamics: Neural Flux Learning for Traffic Forecasting.
Zhe Fu, Benedetto Piccoli, Alexandre M. Bayen.

J12. Field Validation of Kernel-Based Traffic Smoothing Control in Mixed-Autonomy Vehicle Platoons.
Zhe Fu, Xuanmian He, Nicole Han, Bhavya Mamnani, Zihe Liu, Jordan Kam, Alexandre M. Bayen.

J13. TraceFusion: Evidence-Grounded Explanations of CAV Flow-Smoothing Controller Behavior.
Xuanmian He, **Zhe Fu**, Alexandre M. Bayen.

Refereed Conference Proceedings (Published & Under review) * Equal first author, † Corresponding author

C1. "(U)NFV: Supervised and Unsupervised Neural Finite Volume Methods for Solving Hyperbolic PDEs." [link]
The Fourteenth International Conference on Learning Representations (ICLR), 2026.
Zhe Fu*, Nathan Lichtlè*, Alexi Canesse*, Hossein N.Z. Matin*, Maria L.D. Monache, Alexandre M. Bayen.

C2. "Unsupervised Anomaly Detection in Multi-Agent Trajectory Prediction via Transformer-Based Models."
IEEE Intelligent Vehicles Symposium (IV), 2026.
Qing Lyu, **Zhe Fu†**, Alexandre M. Bayen.

C3. "Discovering Safety-Critical Driving Scenarios via Prediction-Based Anomaly Detection."
Transportation Research Board (TRB) Annual Meeting, 2026.
Qing Lyu, **Zhe Fu†**, Alexandre M. Bayen.

C4. "Position and Speed Estimation Using Deep Learning-based KKL Observer in Mixed Autonomy Traffic Systems." [link]
IEEE Conference on Decision and Control (CDC), 2025. **[Oral presentation]**
Yasmine Marani, **Zhe Fu**, Ibrahima N'Doye, Eric Feron, Taous-Meriem Laleg-Kirati, Alexandre M. Bayen.

C5. "Neural Network-Based Solvers for Hyperbolic Conservation Laws: Supervised vs. Unsupervised Learning, and Applications to Traffic Modeling" [link]
2nd International Conference on Neuro-symbolic Systems (NeuS), 2025. **[Oral presentation]**
Zhe Fu*, Alexi Canesse*, Nathan Lichtlè*, Hossein N.Z. Matin*, Zihe Liu, Maria L.D. Monache, Alexandre M. Bayen.

C6. "Pareto Control Barrier Function for Inner Safe Set Maximization Under Input Constraints." [link]
American Control Conference (ACC), 2025. **[Oral presentation]**
Xiaoyang Cao, **Zhe Fu†**, Alexandre M. Bayen.

C7. "Towards Understanding Worldwide Cross-cultural Differences in Implicit Driving Cues: Review, Comparative Analysis, and Research Roadmap." [link]
International conference on intelligent transportation systems (ITSC), 2024. **[Oral presentation]**
Yongqi Dong, Chang Liu, Yiyun Wang, and **Zhe Fu**.

C8. "Kernel-based Planning and Imitation Learning Control for Flow Smoothing in Mixed Autonomy Traffic." [link]
25th *International Symposium on Transportation and Traffic Theory (ISTTT25)*, 2024.
Zhe Fu†, Arwa Alanqary, Abdul Rahman Kreidieh, Alexandre M. Bayen.

C9. "Hierarchical Reinforcement Learning for Scalable Lagrangian Variable Speed Limit in Mixed Autonomy Traffic."
Transportation Research Board 103th Annual Meeting, 2024.
Han Wang, Jonathan W. Lee, **Zhe Fu**, Sharon Hornstein, Daniel Urieli, Maria Laura Delle Monache.

C10. "Cooperative Driving for Speed Harmonization in Mixed-Traffic Environments." [link]
IEEE Intelligent Vehicles Symposium (IV), 2023.
Zhe Fu†, Abdul Rahman Kreidieh, Han Wang, Jonathan W. Lee, Maria Laura Delle Monache, Alexandre M. Bayen.

C11. "A Feedback Control Strategy for Traffic Flow Harmonization in Mixed-traffic Environments."
Transportation Research Board 102th Annual Meeting, 2023.
Zhe Fu†, Abdul Rahman Kreidieh, Alexandre M. Bayen.

C12. "Learning Energy-efficient Driving Behaviors by Imitating Experts." [link]
International conference on intelligent transportation systems (ITSC), 2022. **[Oral presentation]**
Abdul Rahman Kreidieh, **Zhe Fu**, Alexandre M. Bayen.

Refereed Conference Proceeding (Under Review)

C13. Who Bears the Burden? Learning Responsibility for Shared Constraints in Multi-Agent Reinforcement Learning. [link]
Xiaoyang Cao, Jingqi Li, **Zhe Fu**, Alexandre M. Bayen.

Other Articles

O1. "Massive CAV Experiment in Nashville Pits Machine Learning against Traffic Jams." [link]
OR/MS Today, 2023.
Zhe Fu, Kara Manke, Alexandre M. Bayen.

O2. "Streamlining CAV Test Data Collection and Evaluation in the Hardware-in-the-Loop Environment." [link]
UC-ITS-2020-23, 2020.
Zhe Fu, Hao Liu, Xiao-Yun Lu.

REFEREE ACTIVITIES

Editorial Board Member

- Transportation Engineering

Journal Article Reviewer

- ACM Transactions on Knowledge Discovery from Data
- IEEE Open Journal of Intelligent Transportation Systems (OJ-ITS)
- IEEE Transactions on Automatic Control (TAC)
- IEEE Transactions on Intelligent Transportation Systems (T-ITS)
- IEEE Transactions on Vehicular Technology (TVT)
- IEEE Intelligent Transportation Systems Magazine (ITSM)
- Nature Partner Journals (npj) Discover Artificial Intelligence
- Nature Partner Journals (npj) Sustainable Mobility and Transport
- Scientific Reports
- Springer Nature Discover Vehicles
- Transportation Research Record
- Transportation Research Part C: Emerging Technologies
- Transportation Research Part D: Transport and Environment
- Transportation Science

Conference Paper Reviewer

- 25th International Symposium on Transportation and Traffic Theory (ISTTT25)
- ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS)
- IEEE American Control Conference (ACC)
- IEEE Conference on Decision and Control (CDC)
- IEEE European Control Conference (ECC)
- IEEE International Conference on Intelligent Transportation Systems (ITSC)
- IEEE Intelligent Vehicles Symposium (IV)
- Transportation Research Board (TRB) Annual Meeting
- World Transport Convention

Proposal Reviewer

- ITS Berkeley proposals. Transportation Equity Review Committee for ITS Berkeley STRP.

INVITED TALKS & SEMINARS

- Seminar talk, ***CE-CERT***, *UC Riverside, Riverside, CA, Oct. 2026* (link)
- Seminar talk, ***LBNL***, *Berkeley, CA, July 2026*
- Seminar talk, ***ASU***, *Tempe, AZ, April 2026*
- Guest lecture, ***CMU***, *Pittsburgh, PA, April 2026*
- Panelist, ***Agents, Memory, and Simulation***, *Google & SESE, Berkeley, CA, Nov. 2025* (link)
- Guest lecture & Seminar talk, ***UCLA Mobility Lab***, *Los Angeles, CA, Nov. 2025*
- Seminar talk, ***USC AAI Seminar***, *USC, Los Angeles, CA, Oct. 2025*
- Seminar talk, ***eMERGE Seminar***, *UC Berkeley, Berkeley, CA, Sept. 2025* (link)
- Award remark, ***EDGE in Tech Athena Award Ceremony***, *Berkeley, CA, May 2025*
  "Stay Safe, Stay Humble – But Speak Up" , (link)
- Invited talk, ***Mobility-X Lab***, *Rice U, Houston, TX, May 2025*
- Invited talk, ***JTL Urban Mobility Lab***, *MIT, Boston, MA, April 2025* (link)
- Seminar talk, ***Harvard EE Seminar***, *Harvard, Boston, MA, April 2025* (link)
- Invited talk, ***Zardini Lab***, *MIT, Boston, MA, April 2025*

PRESENTATIONS

- Invited session talk, ***INFORMS Annual Meeting 2026***, *San Francisco, CA, Nov. 2026*
- Poster presentation, ***IEEE IV 2026***, *Detroit, MI, June 2026*
- Poster presentation, ***L4DC 2026***, *LA, CA, June 2026*
- Poster presentation, ***ICLR 2026***, *Rio de Janeiro, Brazil, April 2026*
- Poster presentation, ***AI & PDE workshop at ICLR 2026***, *Rio de Janeiro, Brazil, April 2026*
- Oral presentation, ***NorCal Control Workshop***, *Monterey Bay, CA, April 2026* (link)
- Lightning talk, ***COTA TRB Winter Symposium***, *Washington DC, Jan. 2026*
- Poster presentation, ***TRB Annual Meeting 2026***, *Washington DC, Jan. 2026*
- Poster presentation, ***NeurIPS 2025***, *San Diego, CA, Dec. 2025*
- Spotlight talk, ***BARS 2025***, *Stanford, CA, Nov. 2025*
- Poster presentation, ***MIT EECS Rising Stars Workshop***, *MIT, Boston, MA, Oct. 2025* (link)
- Job Market Showcase talk, ***INFORMS Annual Meeting 2025***, *Atlanta, GA, Oct. 2025* (link)
- Oral presentation, ***ACC 2025***, *Denver, CO, July 2025*
- Invited poster presentation, ***ACC 2025 Workshop W05***, *Denver, CO, July 2025* (link)
- Oral presentation, ***NeuS 2025***, *Philadelphia, PA, May 2025*
- Grad Slam Competition, ***Berkeley Grad Slam***, *UC Berkeley, Berkeley, CA, April 2025*
  "Stop-and-Go No More: How a Few Smart Cars Can Fix Traffic Jams" (link), ***Runner-up Winner*** *Campus-wide*
- Poster talk, ***NSF CPS Rising Stars Workshop 2025***, *Nashville, TN, March 2025*
- Session talk, ***POMSHK 2025***, *Hong Kong, China, Jan. 2025*
- Session talk, ***HKSTS 2024***, *Hong Kong, China, Dec. 2024*
- Spotlight talk, ***BARS 2024***, *UC Berkeley, Oct. 2024*
- Poster talk, ***ISTTT25***, *Ann Arbor, MI, July 2024*
- Invited session talk, ***INFORMS Annual Meeting 2023***, *Phoenix, AZ, Oct. 2023*
- Poster competition, ***INFORMS Annual Meeting 2023***, *Phoenix, AZ, Oct. 2023*
  "Cooperative Connected Automated Vehicle Control: Strategies For Speed Harmonization In Mixed Autonomy Traffic.", ***First Place Winner*** *of Multidisciplinary Application Track*
- Interactive poster presentation, ***IEEE IV 2023***, *Anchorage, AK, June 2023*
- Invited session talk, ***IEEE IV Workshop***, *Anchorage, AK, June 2023*
- Poster presentation, ***Berkeley Deep Drive Symposium***, *March 2023*
- Poster presentation, ***TRB Annual Meeting 2023***, *Washington DC, Jan. 2023*
- Spotlight speak & Poster presentation, ***BARS 2022***, *UC Berkeley, Nov. 2022*
- Poster presentation, ***ITS 75 Celebration***, *UC Berkeley, Oct. 2022*

MEDIA & PRESS

- **CIRCLES work featured in Press**, Fortune (12/22, 11/22); AP News (11/22); Popular Science (11/22); PR Newswire (11/22); TechXplore (11/22); UC Berkeley News (11/22); Vanderbilt University News (11/22); Fox Channel 2 KTVU (11/22); Fox Channel 17 WZTV Nashville (11/22); ABC 7 News (11/22); WKRN News Channel 2 (11/22, 11/22); WABI News Channel 5 (11/22); WTVF News Channel 5 (11/22); WSMV Channel 4 (11/22); Main Street Media of Tennessee (11/22); Motor (11/22)
- **1st Place in H2H8 Research Presentation Competition**, ITS Berkeley (8/26)
- **Bridgette Beato Leadership Legacy Graduate Scholarship**, ITS Berkeley (8/26)
- **Outstanding PhD Student of the Year**, ITS Berkeley (5/26)
- **Outstanding Graduate Peer Mentor Award**, Website Announcement
- **CTF Lipman Family Sustainable Transportation Gradaute Fellowship**, ITS Berkeley (3/26)
- **Stanford Energy Postdoctoral Fellowship**, Stanford Energy (2/26)
- **2025 MIT EECS Rising Star**, Website Announcement
- **2025 ENO Fellow**, Eno Center for Transportation (6/25, 5/25); UC Berkeley CEE (6/25); ITS Berkeley (4/25)
- **Athena Award for Graduate Student Leadership**, Website Announcement; CITRIS (5/25); ITS Berkeley (5/25)

- **2nd Place Winner of 2025 Berkeley Grad Slam**, UC Berkeley Graduate Division (4/25); ITS Berkeley (4/25); UC Berkeley CEE (6/25)
- **NSF CPS Rising Star**, Website Announcement
- **CMU ME Rising Star**, UC Berkeley CEE (10/24); ITS Berkeley (10/24)
- **AAa/e Scholarship**, UC Berkeley CEE (10/24); ITS Berkeley (10/24)
- **IEEE ITSC 2024 Institutional Lead Award**, UC Berkeley CEE (10/24); UC Berkeley EECS (8/24); ITS Berkeley (9/24)
- **Dean's Award for Inclusive Excellence**(Oski Student Leadership Award), UC Berkeley Division of Student Affairs (4/24); UC Berkeley CEE (4/24)
- **Honorable Mention of 2023 OR/MS Tomorrow Mini-Poster Competition**, Website Announcement
- **1st Place Winner of 2023 INFORMS Best Poster Competition**, UC Berkeley CEE (2/24); ITS Berkeley (1/24)
- **ITSCA & CTF Scholarship**, UC Berkeley CEE (8/23); ITS Berkeley (8/23)
- **H2H8 Research Grant**, ITS Berkeley (8/23)
- **Outstanding GSI**, Website Announcement; UC Berkeley CEE (4/22); ITS Berkeley (5/22)

OTHER AWARDS & HONORS

Other Awards:

- IEEE ITSC 2024 Institutional Lead Award - CIRCLES Consortium — Sept. 2024
- Honorable Mention, 2023 OR/MS Tomorrow Mini-poster Competition (PhD Category) — Nov. 2023
- **Best Design Presentation**, 2018 BMW Next Mobility Youth Camp — Aug. 2018
- Second Prize, 12th National Transportation Technology Competition (Team Leader, Rank: 9/302) — May 2017
- **First Prize**, Transportation Technology Competition at Tongji University (Team Leader, Rank: 2/59) — April 2017
- Meritorious Winner, US Mathematical Contest in Modeling (Team Leader, Top 7%) — April 2017

Other Scholarships:

- Professional Development Award, *UC Berkeley* — Aug. 2024 & 2025
- Department Award - EECS, *UC Berkeley* — 2023
- Block Grant Award - CEE, *UC Berkeley* — 2019 - 2023
- Summer Department Award - ITS, *UC Berkeley* — May 2021
- Runner-up "Tingya" Outstanding Student Scholarship Competition, *Tongji University* — May 2017

REFERENCES

Dr. Alexandre M. Bayen, Liao Cho Professor (Ph.D. Thesis Advisor) — **Email:** bayen@berkeley.edu
Associate Provost - Berkeley Space Center
Director - CITRIS and the Banatao Institute
Department of Electrical Engineering and Computer Sciences
Department of Civil and Environmental Engineering
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